



UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering (CSE)
Course Syllabus

1	Course Title	Fundamentals of Calculus								
2	Course Code	Math - 1151								
3	Trimester and Year	Spring- 2025								
4	Pre-requisites	N/A								
5	Credit Hours	3								
6	Section	B, C								
7	Class Hours	<u>Section B:</u> Saturday & Tuesday: 09:51 AM - 11:10 AM <u>Section C:</u> Sunday & Wednesday 09:51 AM - 11:10 AM								
8	Class Room	<u>Section B:</u> Room #0708 (7 th Floor) <u>Section C:</u> Room #0702 (7 th Floor)								
9	Instructor's Name	Nasrina Parvin(NPn)								
10	Email	nasrina@ins.uiu.ac.bd , Mobile – 01521102173, Room No.619(A)								
11	Office	Room No # 0619 (A)								
12	Counseling Hours		<table><tr><th>DAY</th><th>TIME</th></tr><tr><td>Saturday & Tuesday</td><td>8:30 AM to 11:10 AM 11:11 AM to 12:30PM</td></tr><tr><td>Sunday & Wednesday</td><td>8:30 AM to 11:10 AM 11:11 AM to 12:30PM</td></tr></table>	DAY	TIME	Saturday & Tuesday	8:30 AM to 11:10 AM 11:11 AM to 12:30PM	Sunday & Wednesday	8:30 AM to 11:10 AM 11:11 AM to 12:30PM	
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13	Text Book	<i>Calculus 10-th Edition - by Howard Anton, Irl Bivens and Stephen Davis</i>								
14	Reference									
15	Course Contents (approved by UGC)	Function, Domain and Range of a Function, Translation, reflection, compression and stretches of a function. Even and Odd functions, Inverse functions, One to One and many to one function Family of Exponential, logarithmic, sine and cosine function, Limit, continuity and differentiability, Tangent line, Derivative and Chain rule, An overview of area problem Newton's anti-derivative method in finding area, Indefinite integral, fundamental theorem of calculus, Definite integral, Area between two curves, arc length.								
16	Course Outcomes (COs)	<table><tr><th>COs</th><th>Description</th></tr><tr><td>CO1</td><td>Define relation of two variables, interval, function, domain and range of different types of function. Composite, odd and even</td></tr></table>			COs	Description	CO1	Define relation of two variables, interval, function, domain and range of different types of function. Composite, odd and even		
COs	Description									
CO1	Define relation of two variables, interval, function, domain and range of different types of function. Composite, odd and even									

			functions, inverse functions, family of Exponential & logarithmic function functions, Limit, continuity, differentiability and integrability of various functions.
		CO2	Analyze graph of functions (Translations, reflections, Stretch and compress), limits and continuity.
		CO3	Compute differentiations and integrations of functions.
		CO4	Apply differentiation to find the rate of change of functions, velocity of a moving particle, average velocity, instantaneous velocity and slope of secant and tangent line.
		CO5	Interpret the relationship of definite integral with area problem and indefinite integral and arc length.

17 Teaching Methods Lecture, Case Studies, Assignments.

18	CO with Assessment Methods			
		CO	Assessment Method	(%)
		-	Attendance	5
		1-5	Assignments	5
		1-5	Class Tests	20
		×	Presentation	10
		1-2	Midterm exam	30
		3-5	Final exam	40

19 Mapping of COs and Program outcomes

COs	Program Outcomes(POs)											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3	√											
CO4	√											
CO5		√										

20 Lecture Outline

Class	Topics/Assignments	COs	Reading Reference	Lecture Outcomes/ Activities
1-4	Function, Domain and Range of a Function	1	Chap: 0.1 Exercise: 1, 3, 7, 9	Q/A Test, Assignment
5-7	Translation, reflection, Even and Odd function	1, 2	Chap: 0.2 Exercise: 5 – 24	Q/A Test, Assignment

8-10	Inverse function, One to One, many to one function and Family of Exponential & logarithmic function	1, 2	Chap: 0.4 Exercise: 1, 3 Chap: 0.5 Page 53-Figure 0.5.2, Page 54-example-1, Page 56-Figure-0.5.6.	Q/A Test, Assignment
11-12	Limit and Continuity	1, 2	Chap: 1.1 Example-5, 7; Exercise: 1 – 8 Chap: 1.5 Example-2, 3, 4; Exercise-11,13,15,17,19	Q/A Test, Assignment
MID Examination				
13-16	Tangent lines and Rate of change, Differentiation of different types of functions (Derivative function and Chain rule)	1, 3, 4	Chap: 2.1 Exercise:11 – 18 Chap: 2.2 Exercise: 23, 25, 26, 47 & 48 Chap: 2.6 Exercise: 7 – 49 (odd)	Q/A Test, Assignment
17	An overview of area problem, Newton's anti-derivative method in finding area	1, 5	Chap: 5.1 Example: 1, 2; Exercise: 13 – 18	Q/A Test, Assignment
18-22	Indefinite integral, fundamental theorem of calculus, Definite integral and Area between two curves	3, 5	Chap: 5.5 Exercise: 13 – 20 Chap: 5.2 Exercise: 5 – 8, 15 – 30 Chap: 5.3 Exercise: 15 – 56 Chap: 5.6 Exercise:13 – 30 Chap: 6.1 Exercise:1 – 7, 11 – 12	Q/A Test, Assignment
23-24	Different types of Integration (Principles of Integral evaluation, Integration by parts and Trigonometric Substitution).	5	7.1: A review of familiar integration formulas: 1-13, 21-26. Exercise Set 7.1: 1-30 7.2: Examples: 1-8 Exercise Set 7.2: 1-38 7.4: Examples: 1-6 Exercise Set 7.4: 1-26, 37-47 (Only odd numbers)	Q/A Test, Assignment
FINAL Examination				
*** Some Important Dates: 7th class (CT-1), 11th class (CT-2), 19th class (CT-3), 23rd class (CT-4)				

Appendix 1: Assessment Methods

Assessment Types	Marks
Attendance	5%
Assignments	5%
Class Tests	20%
Mid Term	30%

Final Exam	40%
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Appendix 2: Grading Policy

Letter Grade	Marks %	Grade Point	Letter Grade	Marks%	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Appendix-3: Program outcomes

POs	Program Outcomes
PO1	An ability to apply knowledge of mathematics, science, and engineering
PO2	An ability to identify, formulate, and solve engineering problems
PO3	An ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability
PO4	An ability to design and conduct experiments, as well as to analyze and interpret data
PO5	An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice
PO6	The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context
PO7	A knowledge of contemporary issues
PO8	An understanding of professional and ethical responsibility
PO9	An ability to function on multidisciplinary teams
PO10	An ability to communicate effectively
PO11	Project Management and Finance
PO12	A recognition of the need for, and an ability to engage in life-long learning